

Lecture 8: Black Box Methods: Support Vector Machines and Neural Networks

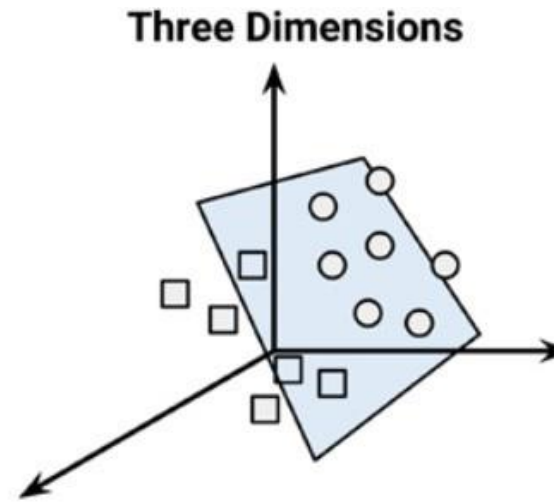
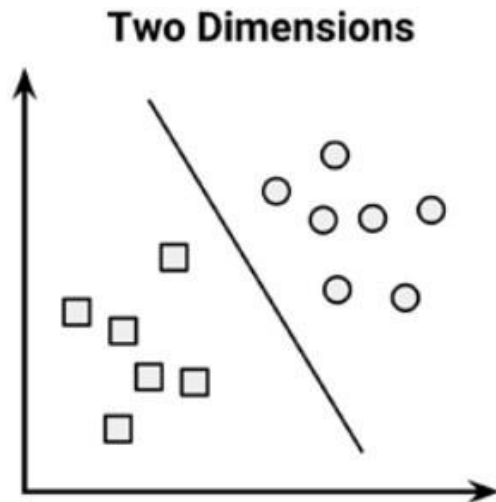
Support Vector Machines

Black Box Methods

- Black box methods refer to a pair of methods that are extremely powerful, while their inner workings can be difficult to understand.
 - The mechanism that transforms the input into the output is obfuscated by an imaginary box.
 - Complex mathematics allowing them to function.
- Although they may not be easy to understand, it is dangerous to apply black box models blindly.

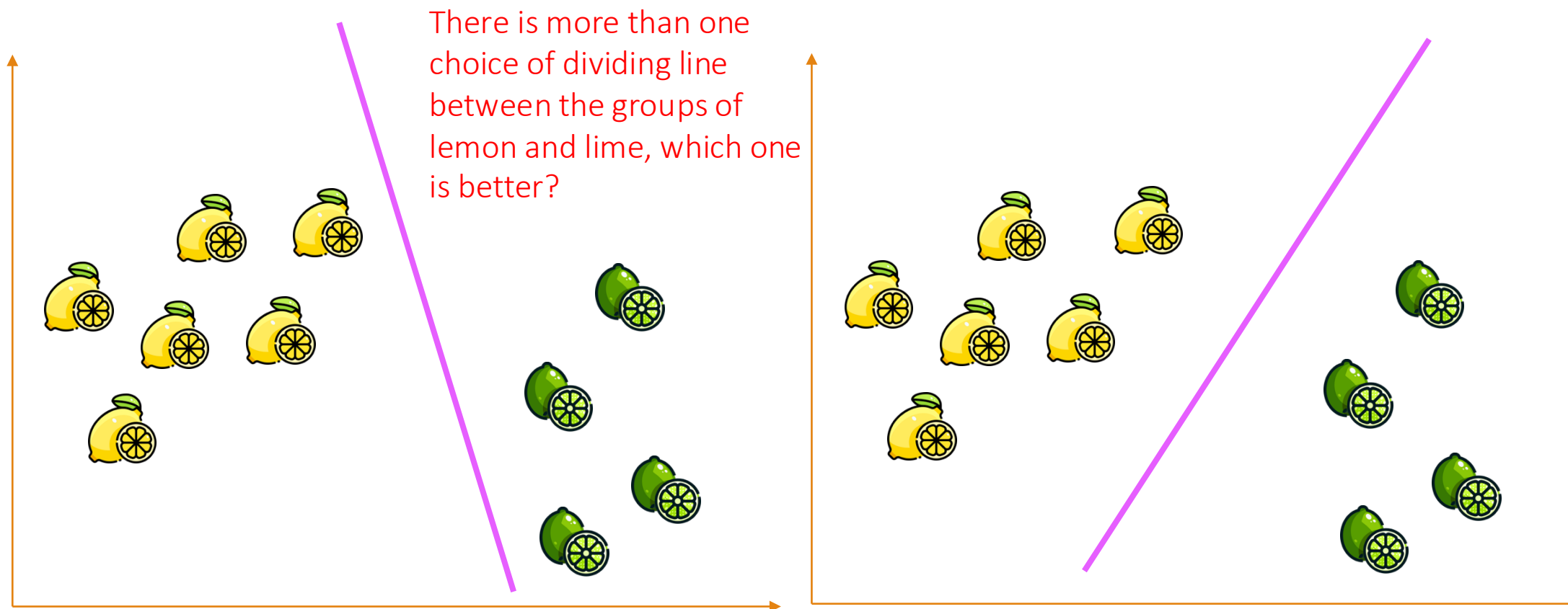
Understanding Support Vector Machines

- A Support Vector Machine (SVM) can be imagined as a surface that creates a boundary between points of data plotted in multidimensional that represent examples and their feature values.
- The goal of a SVM is to create a flat boundary called a hyperplane, which divides the space to create fairly homogeneous partitions on either side.

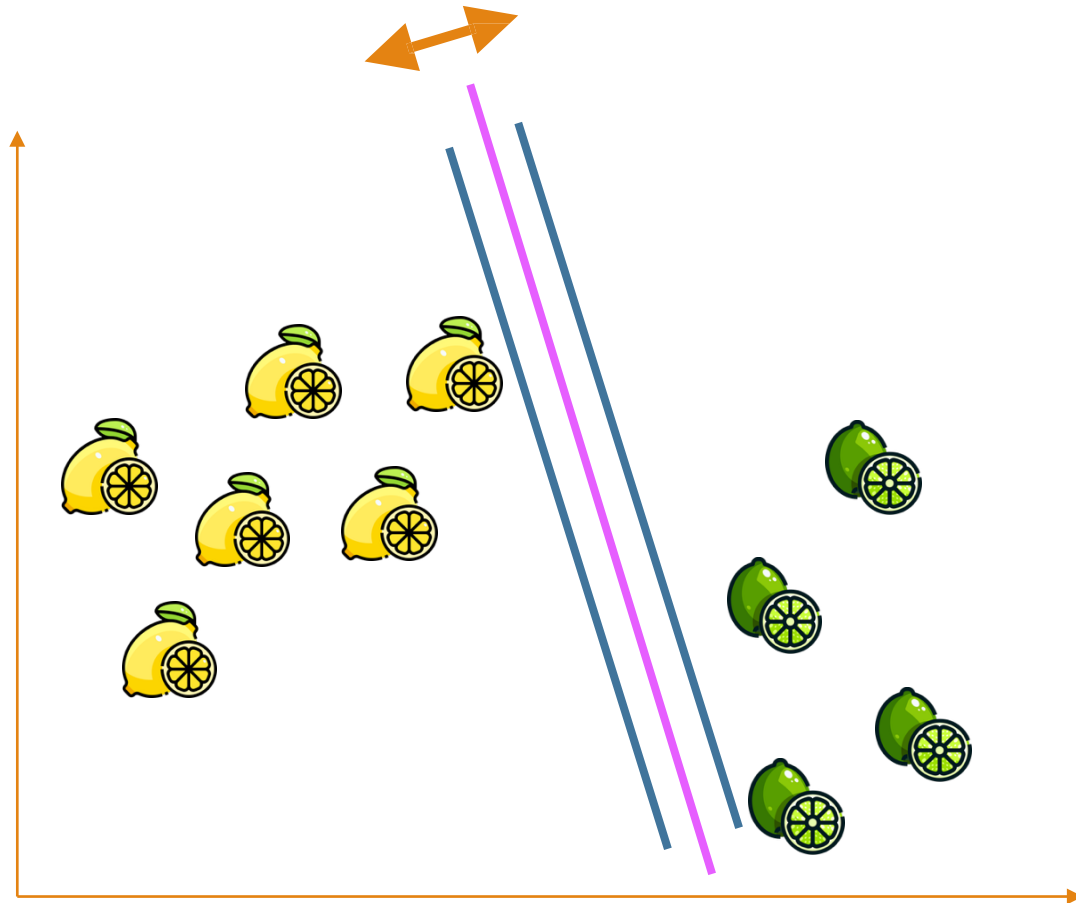


Classification with Hyperplanes

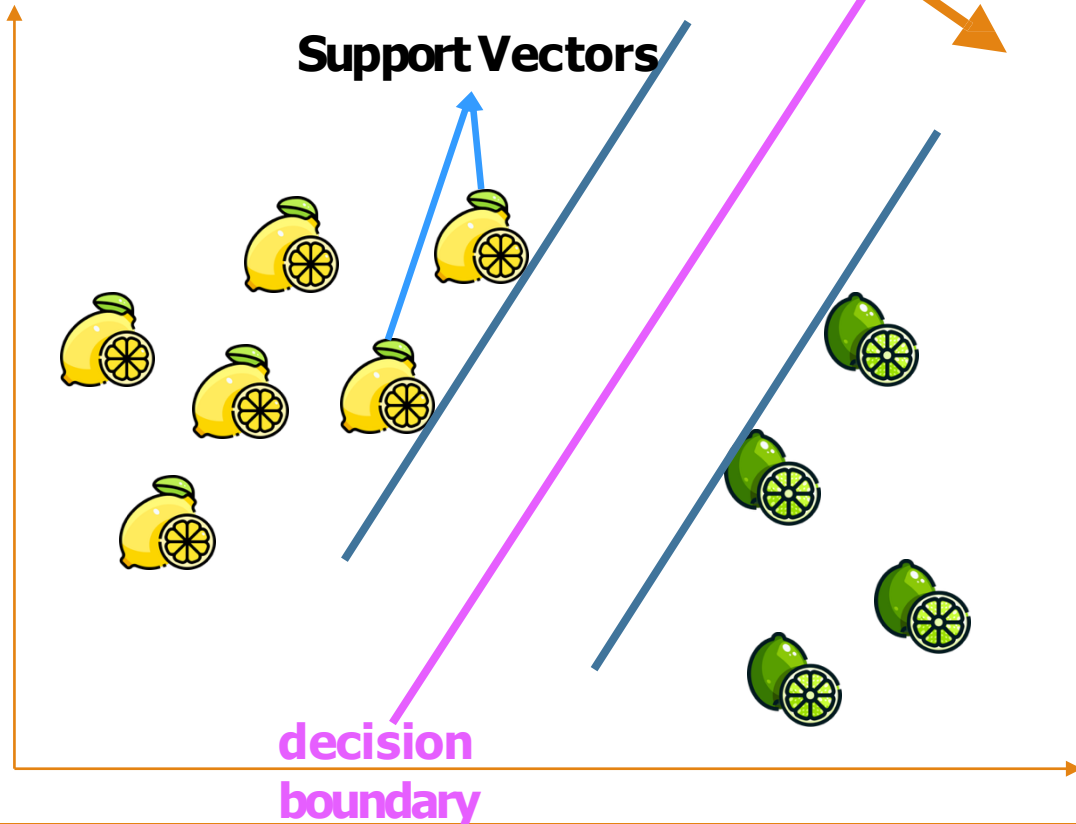
- Suppose we want to classify the lime and lemon



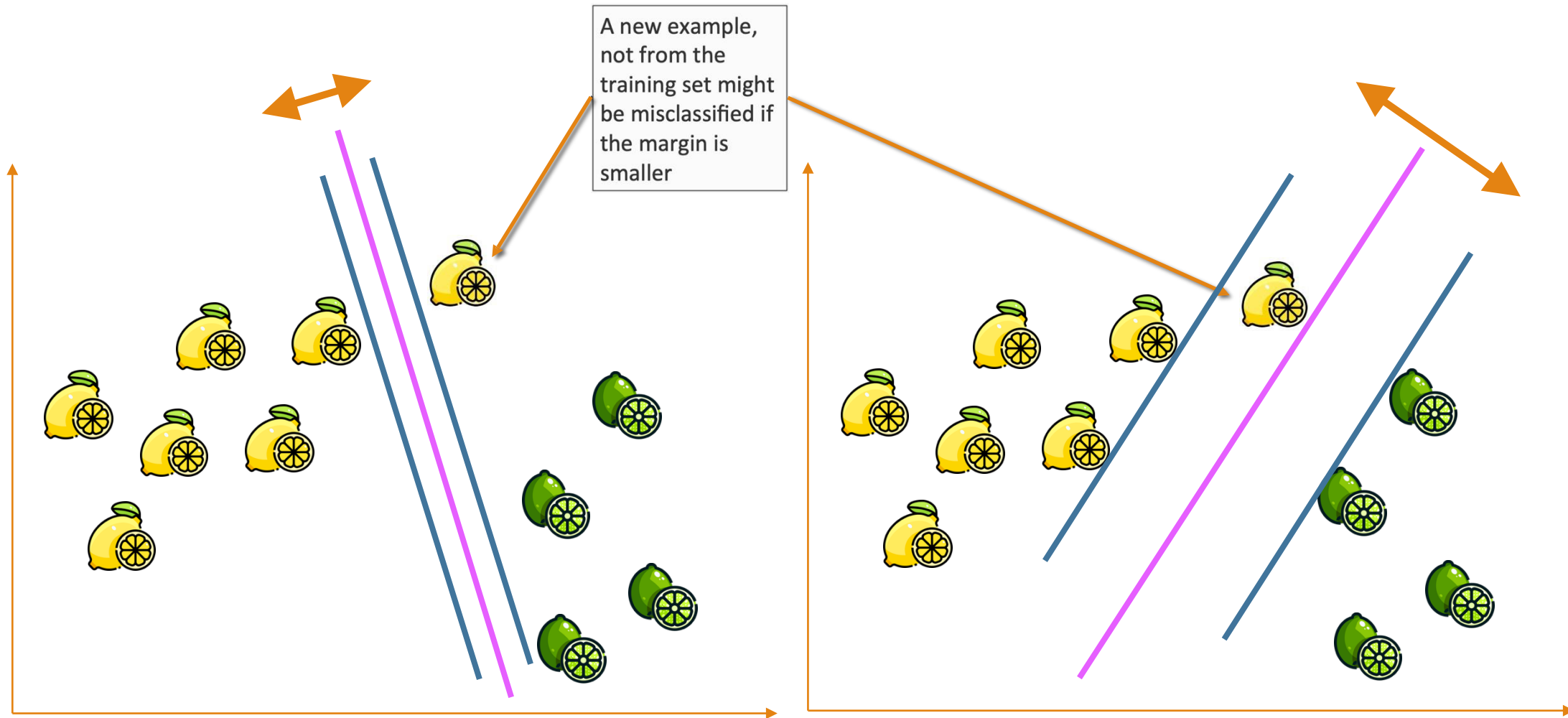
Classification with Hyperplanes



This is the wider margin to create the greater separation between the two groups.

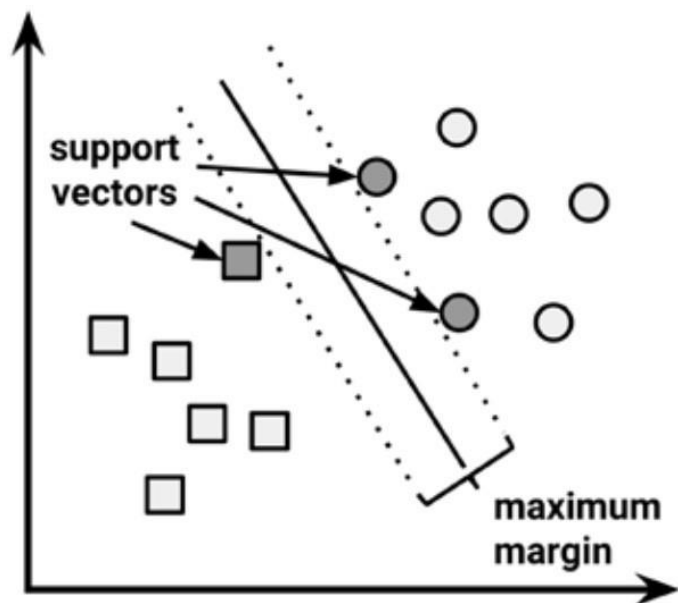


Classification with Hyperplanes



Classification with Hyperplanes

- SVM searches for the **Maximum Margin Hyperplane (MMH)** that creates the greatest separation between the two classes.
- The maximum margin will improve the chance that, in spite of random noise, the points will remain on the correct side of the boundary.

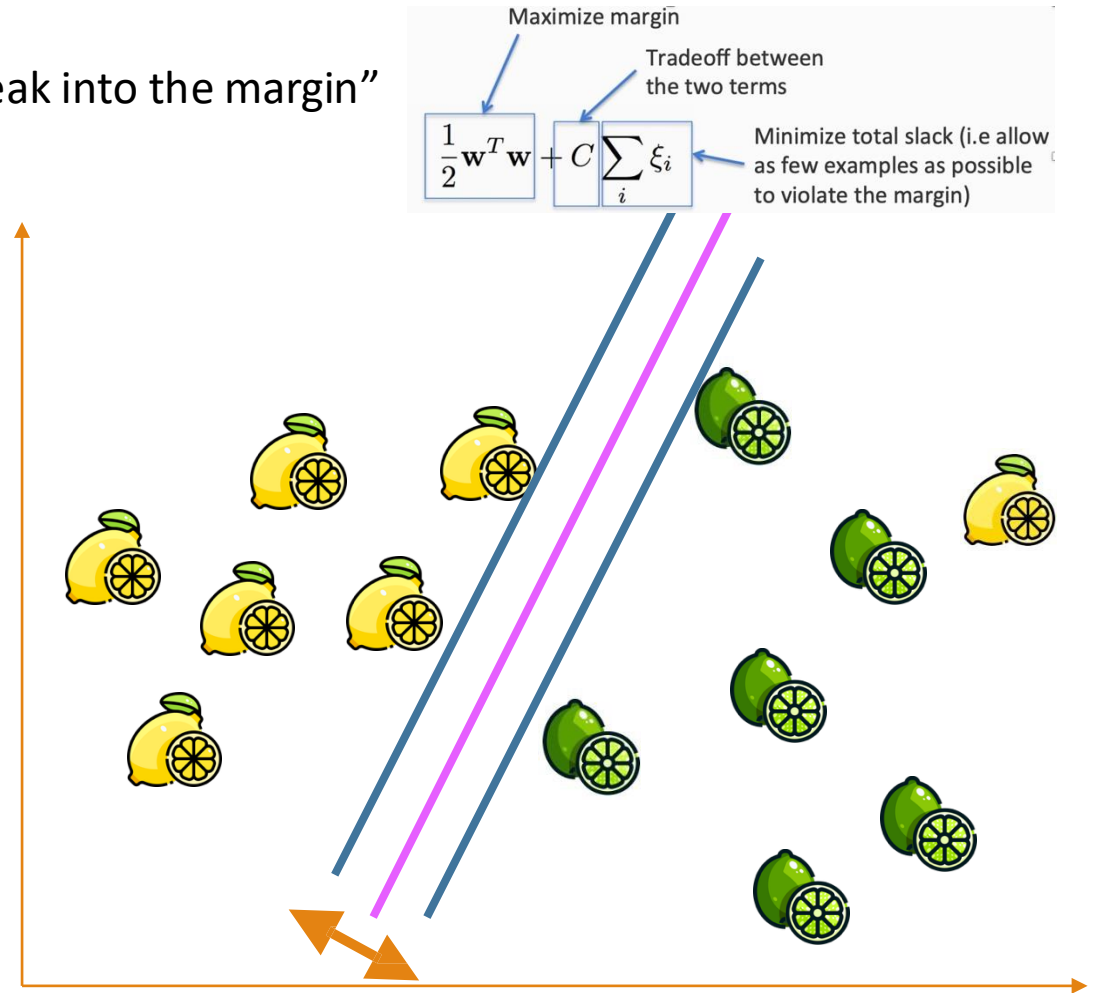
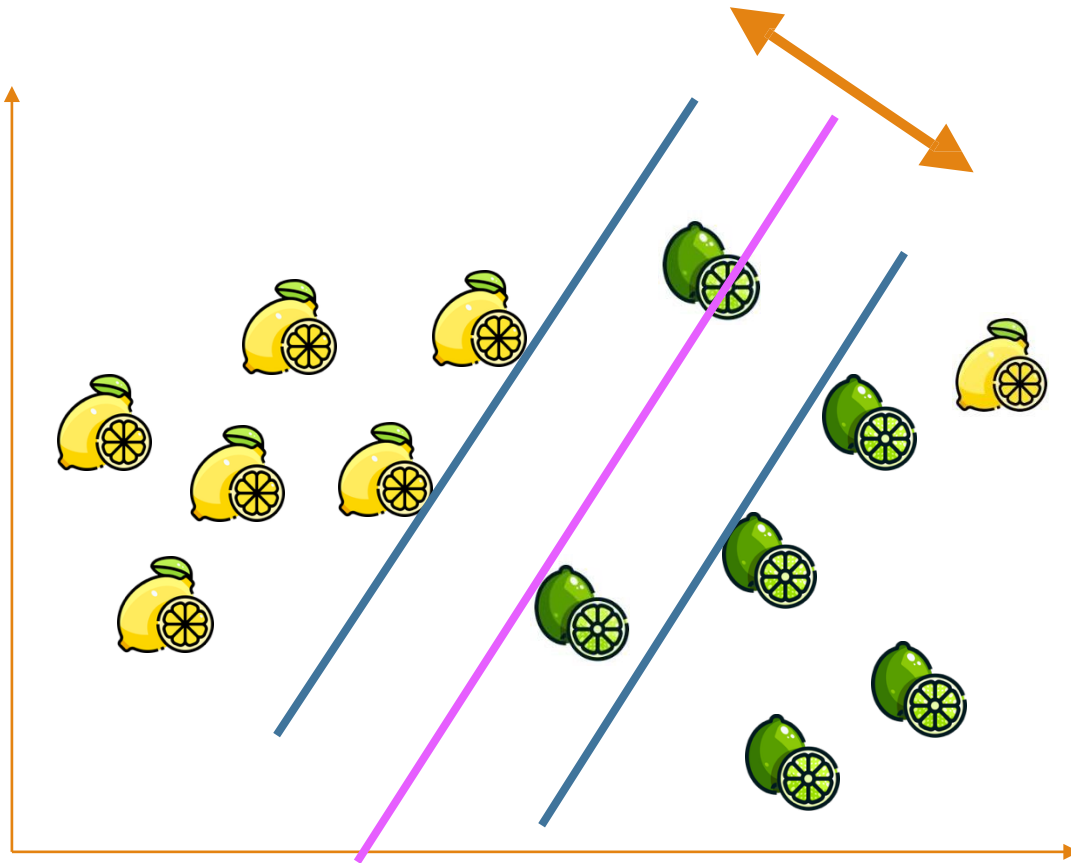


- The support vectors (indicated by arrows in the figure that follows) are the points from each class that are the closest to the MMH
- Each class must have at least one support vector, but it is possible to have more than one.
- Using the support vectors alone, it is possible to define the MMH.
- ***The margin of a hyperplane is the distance between the hyperplane and the data point nearest to it.***

Classification with Hyperplanes

The case of nonlinearly separable data

- A cost value (denoted as C) is applied to all points that “break into the margin”



Maximize margin

Tradeoff between the two terms

$$\frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_i \xi_i$$

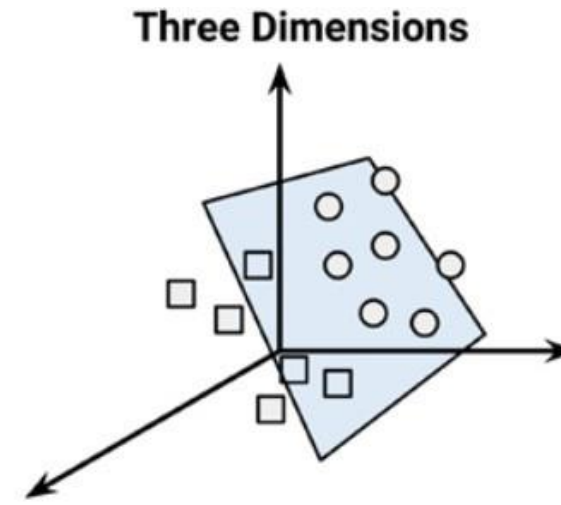
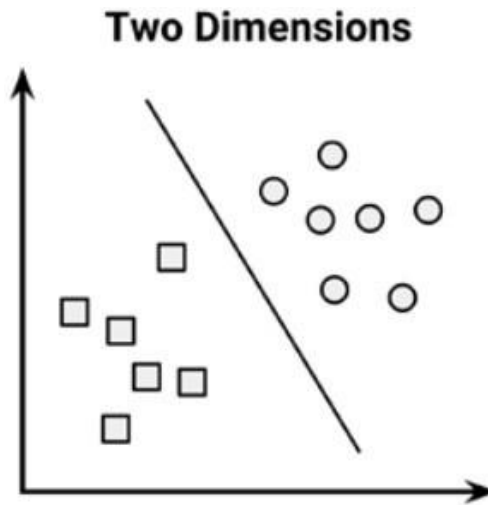
Minimize total slack (i.e. allow as few examples as possible to violate the margin)

Linearly separable

- SVMs can be adapted for use with nearly any type of learning task, including both classification and numeric prediction.
- SVMs use a boundary called a hyperplane to partition data into groups of similar class values.

linearly separable:

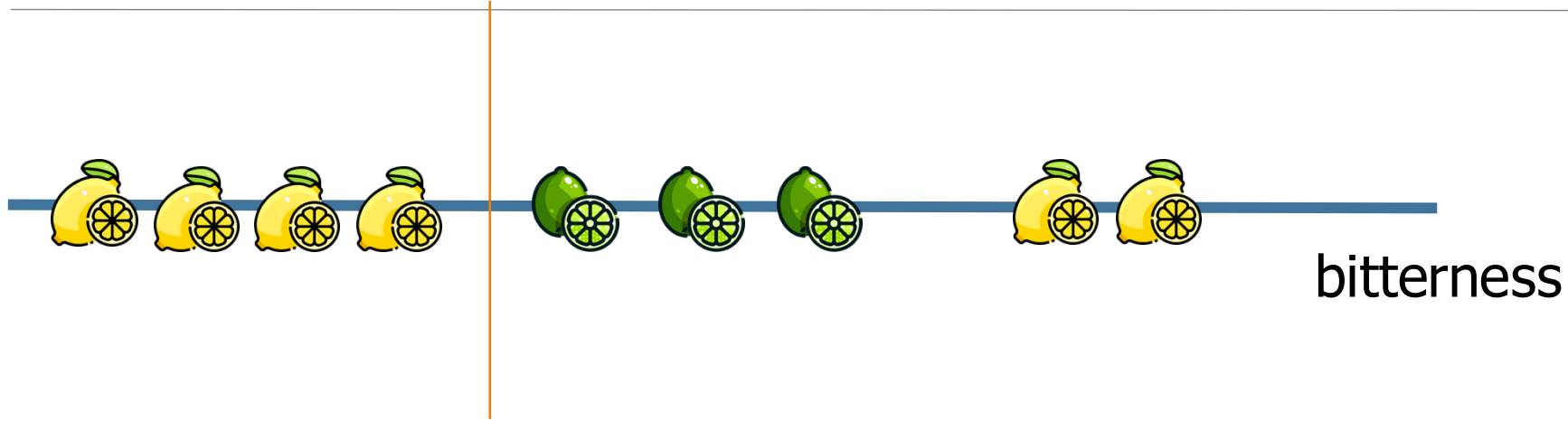
the circles and squares can be separated perfectly by the straight line or flat surface



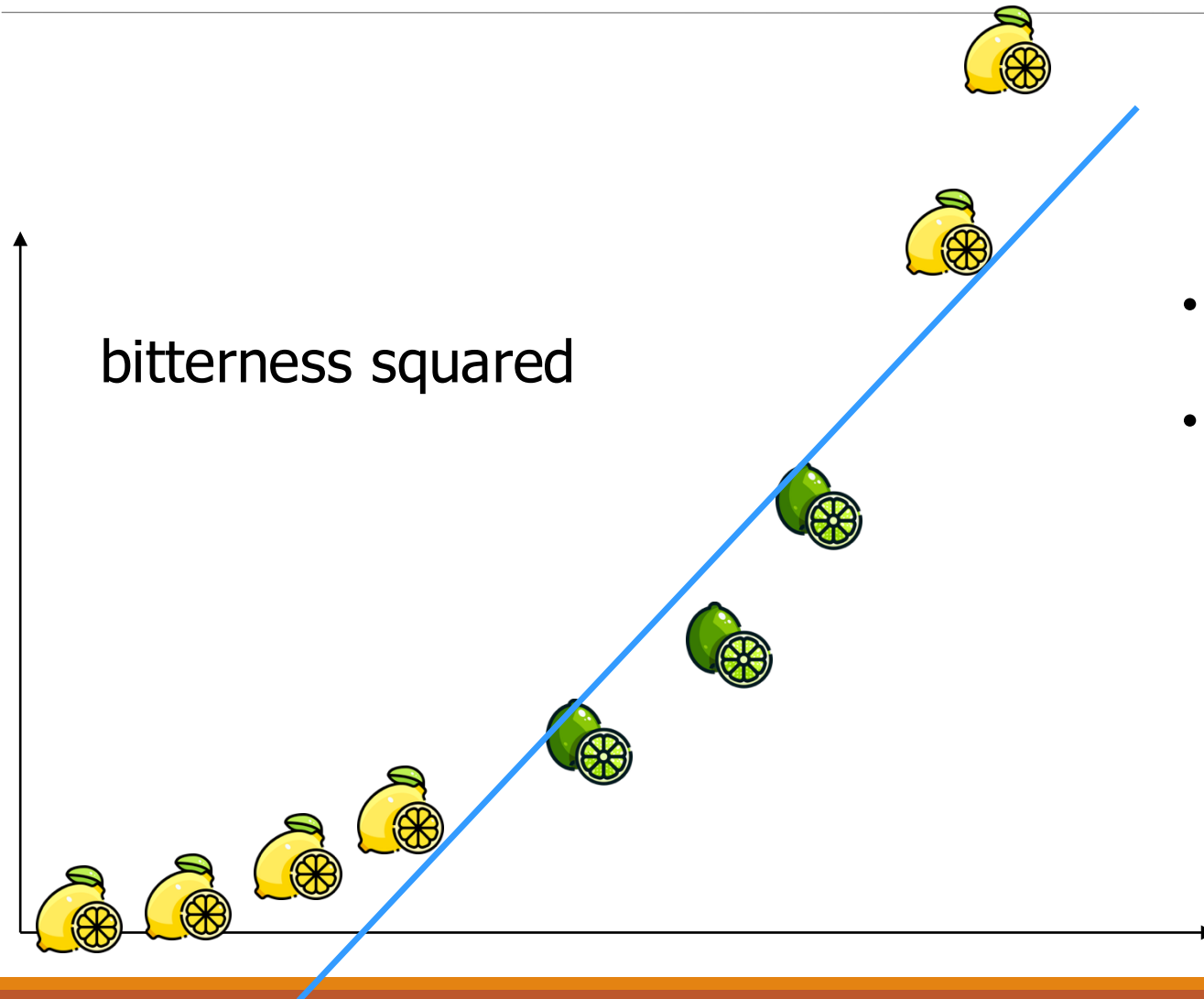
Classification with Hyperplanes

- In many real-world applications, the relationships between variables are nonlinear.
- Another key feature of SVMs is their ability to map the problem into a higher dimension space using a process known as the **kernel trick**.

Using Kernels for Non-linear Spaces

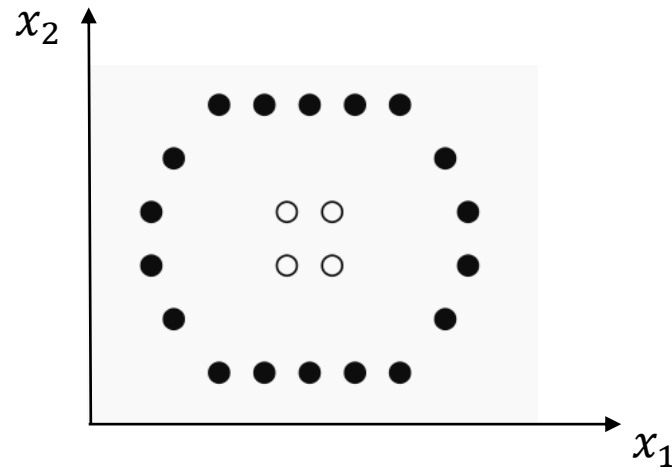


Using Kernels for Non-linear Spaces



- Kernel function decides how we project data points.
- In this case, we use the polynomial Kernel.

Using Kernels for Non-linear Spaces



○ = one class (center)

● = another class (surrounding)

We transform each point:

$$(x_1, x_2) \rightarrow (x_1, x_2, x_1^2 + x_2^2)$$

This adds a **third dimension (z-axis)**.

What happens in 3D?

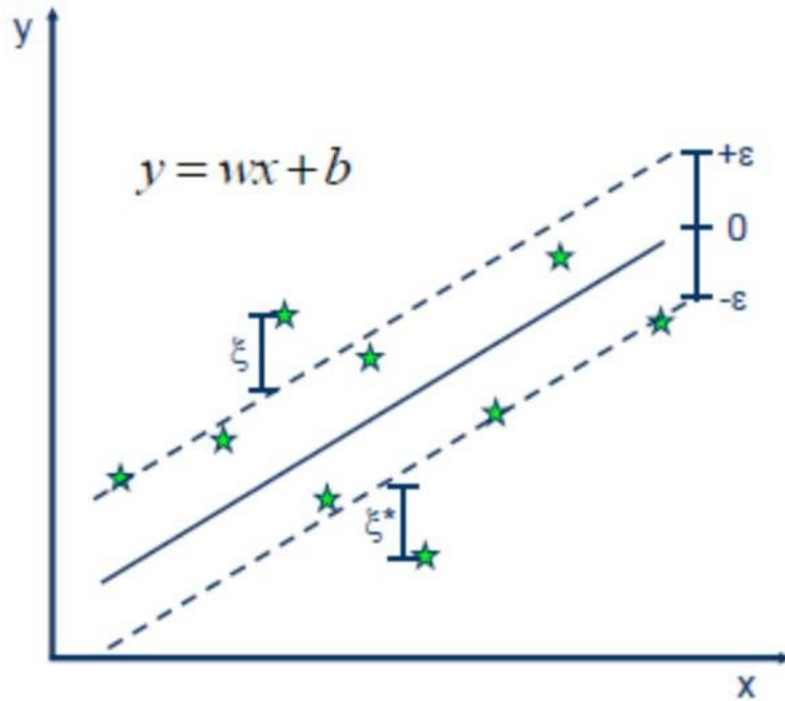
Using Kernels for Non-linear Spaces

- SVMs with nonlinear kernels add additional dimensions to the data in order to create separation in this way.
- The kernel trick involves a process of constructing new features that express mathematical relationships between measured characteristics.
- This allows SVM to learn concepts that were not explicitly measured in the original data.
 - Common kernels functions include the linear kernel, polynomial kernel, sigmoid kernel, and Gaussian RBF kernel.

Support Vector Regression

- The Support Vector Regression (SVR) uses the same principles as the SVM for classification, with only a few minor differences.
 - Output is a real number
 - A margin of tolerance (epsilon) is set in approximation to the SVM
 - To minimize error, individualizing the hyperplane which maximizes the margin

Support Vector Regression



- Minimize:

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^N (\xi_i + \xi_i^*)$$

- Constraints:

$$y_i - wx_i - b \leq \epsilon + \xi_i$$

$$wx_i + b - y_i \leq \epsilon + \xi_i^*$$

$$\xi_i, \xi_i^* \geq 0$$

It also has **kernel trick** to map the data into a higher dimension space.

Evaluation Metrics for Numeric Prediction

- SVM regression with different C values

- `model_SVM = SVR(C = 10.0, kernel = 'poly', degree=5)`

Performance on training data:

MAE: 4839.047438488246
RMSE: 8053.03055999187

Performance on testing data:

MAE: 4671.790852087691
RMSE: 7734.033743681172

- `model_SVM = SVR(C = 1000.0, kernel = 'poly', degree=5)`

Performance on training data:

MAE: 2265.2265388041433
RMSE: 5276.840377345534

Performance on testing data:

MAE: 2356.7151857362214
RMSE: 4954.465367870135

- `model_SVM = SVR(C = 100000.0, kernel = 'poly', degree=5)`

Performance on training data:

MAE: 1763.067797384399
RMSE: 4767.9874432200995

Performance on testing data:

MAE: 2494.1724118464012
RMSE: 5839.895713110879

Which model
has better
performances?

Neural Networks

Understand Neural Networks

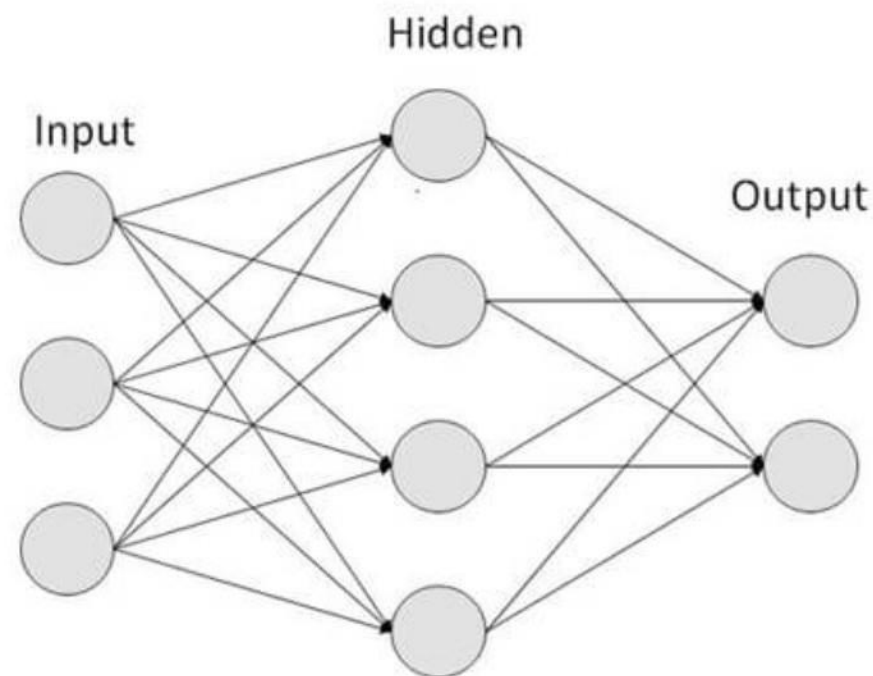
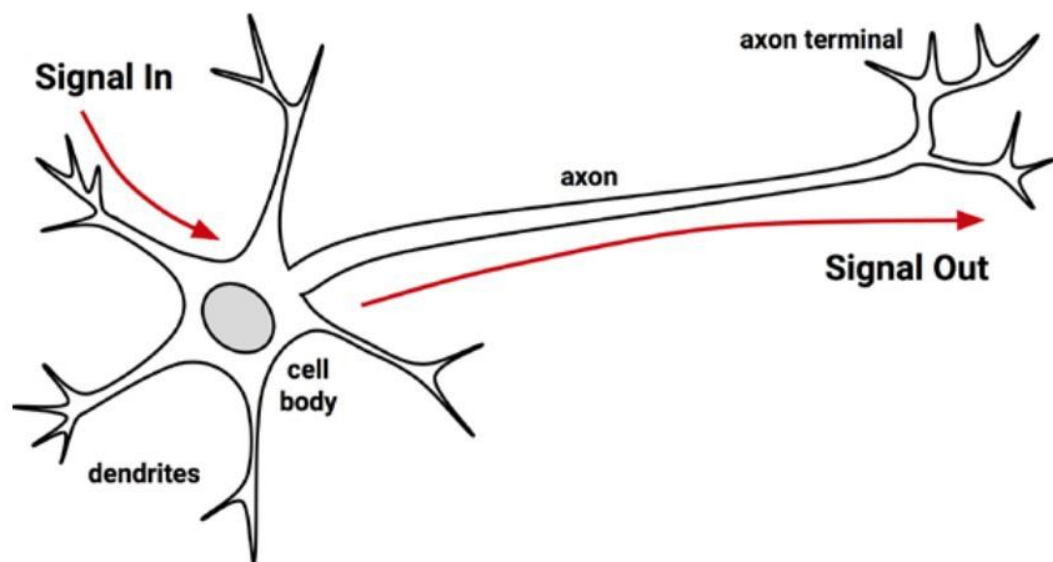
- Neural networks, are computational models inspired by biological neural networks, particularly, they are inspired by the behavior of neurons and the electrical signals they convey between input, processing, and output from the brain.

Understand Neural Networks

- An Artificial Neural Network (ANN) models the relationship between a set of input signals and an output signal
 - Derived from our understanding of how a biological brain responds to stimuli from sensory inputs
 - A brain uses a network of interconnected cells called **neurons** to create a massive parallel processor
 - ANN uses a network of **artificial neurons** or **nodes** to solve learning problems
- ANNs are versatile learners that can be applied to nearly any learning tasks
 - Classification, numeric prediction, and even unsupervised pattern recognition

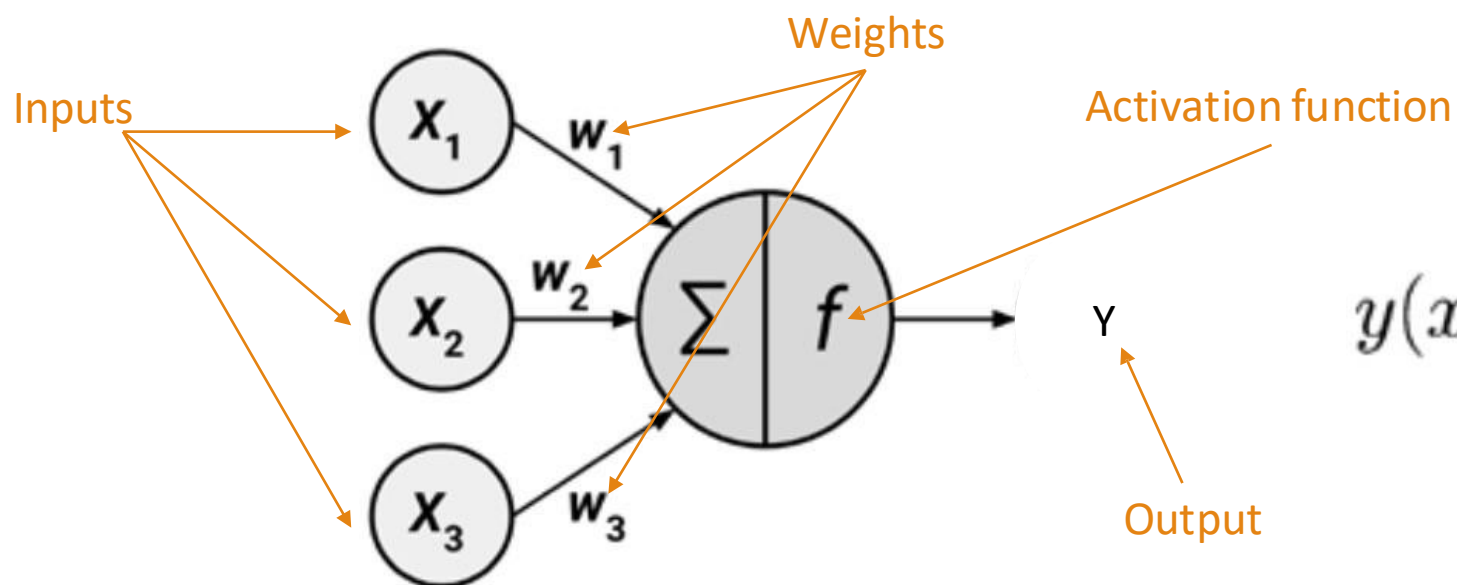
From Biological to Artificial Neurons

- Nature has inspired millions of innovations-the human brain has inspired the creation of neural networks.



From Biological to Artificial Neurons

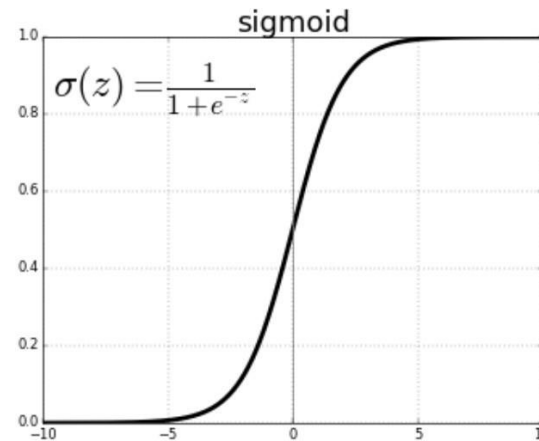
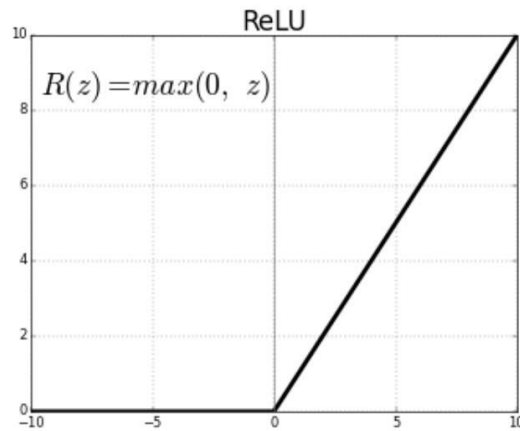
- The model of a single artificial neuron can be understood in terms very similar to the biological model.



$$y(x) = f \left(\sum_{i=1}^n w_i x_i \right)$$

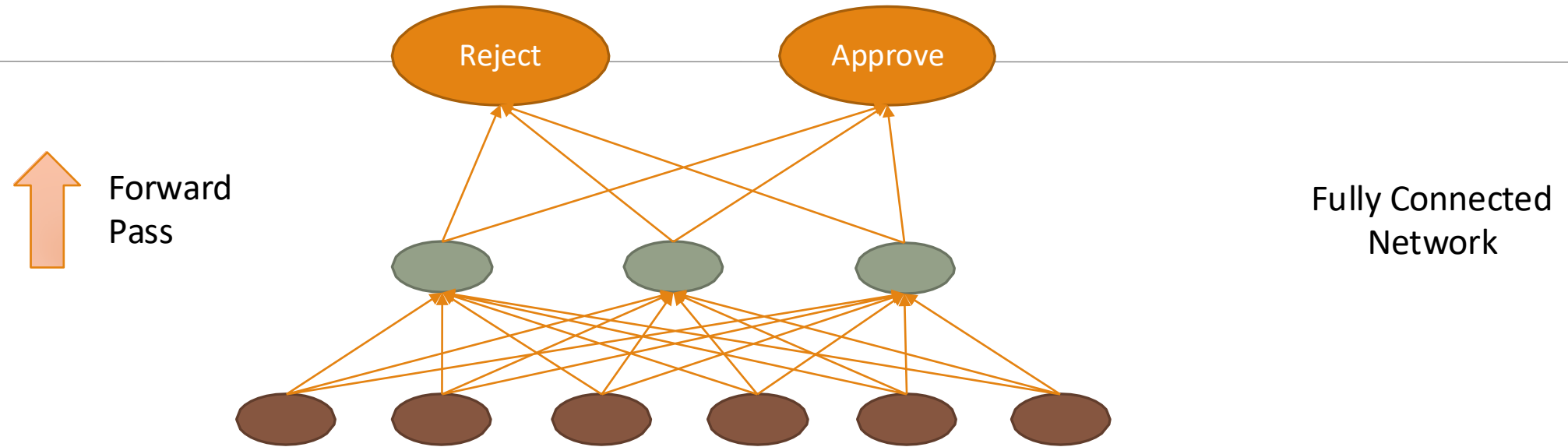
Activation Functions

- Activation function decides, whether a neuron should be activated or not by calculating weighted sum and further adding bias with it.
- The purpose of the activation function is to **introduce non-linearity** into the output of a neuron.



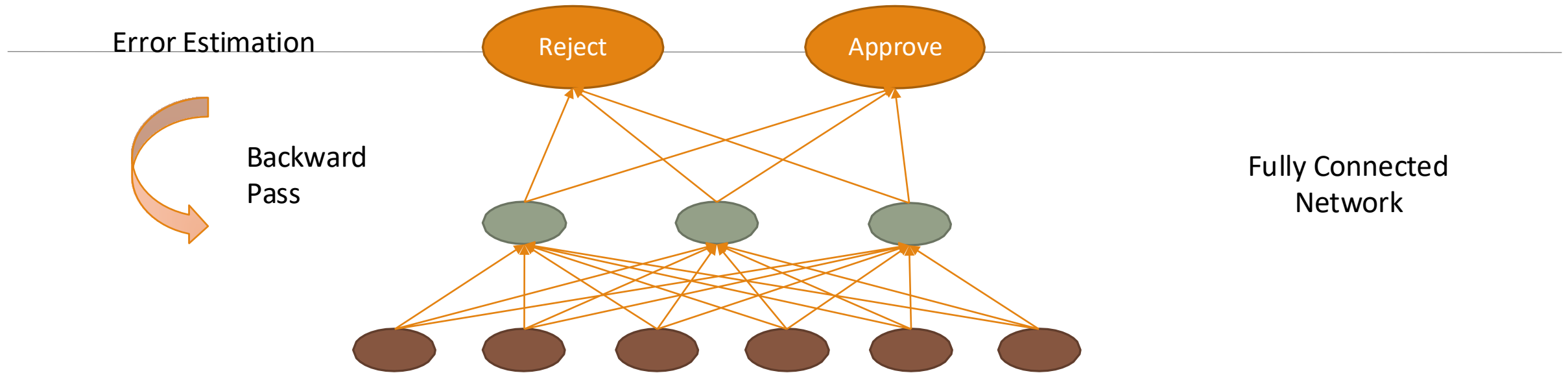
Multilayer Perceptron (MLP)

Neural Network: Forward Pass



CID	Credit Score	Income (10,000)	Term	Age	Gender	Number of Children	Status
1	6	2	1	35	0	0	Reject
2	7.5	5	3	38	0	2	Approve
3	4	3.5	2	23	1	4	Reject
...							
100000	8	2.5	1	35	0	1	Approve

- Epoch: One forward-pass and one backward-pass of all the training sample.



Step 1: Model initialization: Randomly set initialized neural network

Step 2: Calculate the error: How far is your model output from the actual output.

Step 3: Minimize Error: Check whether the error is minimized or not.

Step 4: Update the parameters: Update the parameters (weights and biases) until the error becomes minimum.

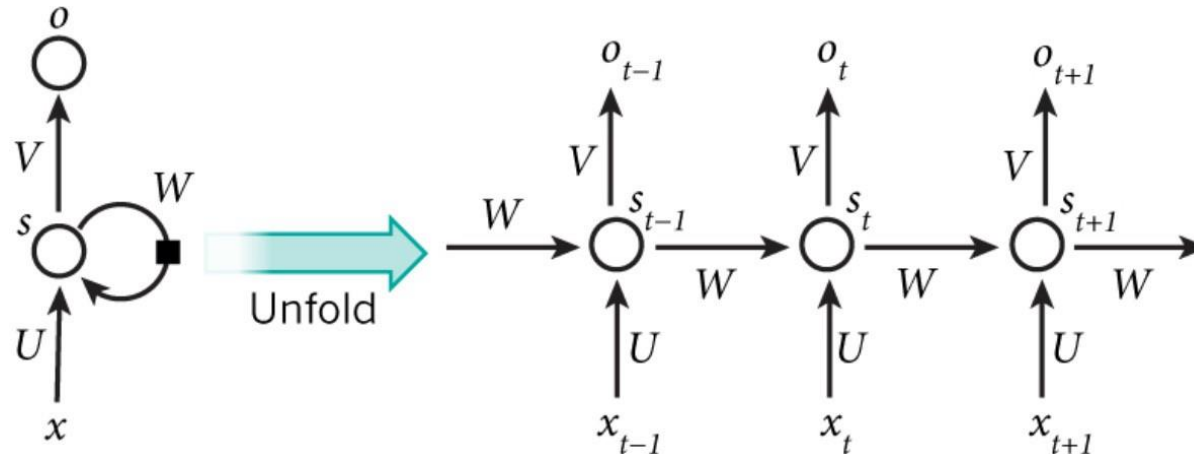
Step 5: Model is ready to make a prediction; Once the error becomes minimum, we can feed some inputs to your model and it will produce the output.

Network Topology

- Generally, larger and more complex (**more layers & more nodes in each layer**) networks are capable of identifying more subtle patterns and complex decision boundaries.
 - The number of layers
 - The number of nodes within each layer of the network
 - A greater number of neurons will result in a model that more closely mirrors the training data, but this runs a risk of **overfitting**. Large neural networks can also be computationally expensive and slow to train. It may generalize poorly to future data.

Network Topology

- In contrast, a **recurrent network** (or **feedback network**) allows signals to travel in both directions using loops.
 - The idea behind RNN is to make use of sequential information.
 - Before we assume all inputs are independent of each other.
 - RNN performs the same task for every element of sequence, with the output being depended on the previous computations.
 - Have the “memory” which captures information about what has been calculated so far.



When Might ANN Be a Good Choice?

- Advantages of deep learning:
 - Scalability (see picture)
 - Automatic feature selection
- Results get better with
 - More data
 - Bigger models
 - More computation

